

Session 12: Data Types Grab Bag — Pen-and-Paper Pair Exercise

PSY 410 | Data Science for Psychology

Name: _____ Date: _____

No laptop today? No problem. This handout lets you practice the same skills on paper. Work with a partner who has a laptop and compare your work at the end.

The data: `attention_data`

You have survey data with a 1–7 attention check item where the correct answer is 4:

participant_id	attention_check
1	4
2	3
3	4
4	7
5	NA
6	4

The task (same as the slide exercise)

1. Create a new column `passed` that is `TRUE` if they answered 4, `FALSE` otherwise
2. Create a column `status` with three values: “Passed”, “Failed”, or “No response” (for NA)
3. What proportion of participants passed?

Your pen-and-paper version

Steps 1 & 2: Fill in `passed` and `status` for each row. `passed` is `TRUE` when `attention_check == 4`. `status` uses `case_when()` logic — check conditions in order (“No response” for NA, “Passed” for 4, “Failed” otherwise).

participant_id	attention_check	passed	status
1	4		
2	3		
3	4		
4	7		

participant_id	attention_check	passed	status
5	NA		
6	4		

Tricky question: What does `NA == 4` return? (Not FALSE!) _____

Why do we need to check for `is.na()` first in `case_when()`?

Step 3: Calculate the proportion who passed. Using `mean(x == 4, na.rm = TRUE)`:

- How many non-NA values are there? _____
- How many of those equal 4? _____
- Proportion passed = _____ / _____ = _____

Step 4: Write the code. Fill in the blanks:

```
attention_data |>
  mutate(
    passed = attention_check _____ 4,
    status = case_when(
      _____(attention_check) ~ "No response",
      attention_check == 4 ~ "_____",
      .default = "_____"
    )
  )

# Proportion who passed
attention_data |>
  summarize(prop_passed = _____(attention_check == 4, na.rm = _____))
```

Check your work

Compare your filled-in tables and code with your partner's screen. Do your answers match?